

Akashdeep Majumder

Frontend Engineer · Real-Time Systems & Robotics · React · Redux · React Three Fiber

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SUMMARY

Frontend-focused Software Engineer with **3+ years of experience** building high-performance, real-time interfaces for robotics systems and full-stack web applications. Deep expertise in **React, Redux Toolkit, and React Three Fiber** — shipped a 3D AMR telemetry visualizer, a Warehouse Management System, and a full-stack social platform from architecture to production. Reaches into FastAPI, Node.js, and WebSockets when the system demands it.

TECHNICAL SKILLS

Frontend: React.js (v18/v19), Next.js (App Router, SSR, Middleware), Redux Toolkit, React Router, React Three Fiber, Three.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, Material UI, Electron.js

Backend: FastAPI, Node.js, Express.js, Django, REST APIs, WebSockets, Socket.IO, Redis, Python

Database: MySQL, MongoDB

Testing: Jest, React Testing Library, Unit Testing

DevOps & Tools: Docker, Git, Vite, CI/CD, Vercel, Render

Robotics: ROS2 (Humble), LiDAR Point Cloud Processing, AMR

Other: WebGL, Immer (undo-redo), Zod

WORK EXPERIENCE

Thedush Robotics *March 2023 – April 2026*

Software Engineer · Coimbatore, India

ONIX – Robot Telemetry & Visualization Software

- Architected a **high-performance 3D visualization pipeline** for AMRs using React 18, Three.js, and React Three Fiber (R3F) to track live robot telemetry — migrated from Canvas 2D when larger warehouses pushed frame rates below smooth thresholds.
- Optimized LiDAR data streaming (JSON -> Float16 + Deflate compression), **reducing bandwidth by ~93% (1.5 MB -> 100 KB)** and boosting render performance **from 5 FPS to 50+ FPS**.
- Implemented an **Immer-based undo-redo system** using patch-based state diffing, **cutting memory usage by ~90%** vs. full-snapshot approach.
- Built a real-time dynamic mapping and collision detection system via WebSockets, FastAPI, and ROS2 (Humble).

Stack: React.js v18, Vite, Three.js, React Three Fiber, Redux Toolkit, WebSockets, Material UI, FastAPI, MySQL, Redis, WebGL

WMS + Robot + PTL Integration System

- Architected and delivered a full **Warehouse Management System (WMS)** integrating Robots and Pick-to-Light (PTL) as a proof of concept for the Indian Army, **in just 14 days**.
- Automated item retrieval with robot navigation + PTL LED guidance, **reducing picking time by ~50%**; replaced manual paper records with a centralized database enabling real-time demand/issue tracking.
- Packaged as **air-gapped offline desktop apps** (Electron .exe + PyInstaller bundle) for LAN-only deployment on factory-floor Windows machines via USB — no internet required.

Stack: React.js v18, Vite, Redux Toolkit, WebSockets, ElectronJS, Material UI, FastAPI, MySQL, ROS2, Modbus TCP

Robot Configuration Builder

- Designed a React-based guided UI to generate deeply nested **600+ line robot configuration JSON files**, replacing error-prone manual editing for deployment engineers.
- Automated JSON validation and generation — **cutting deployment config time by ~50%** and **reducing configuration errors by ~90%**.

Stack: React.js, Vite, Redux Toolkit, Material UI, JavaScript

Network Strength Visualizer

- Built a **3D Wi-Fi heatmap tool** that samples signal strength at each robot position during warehouse navigation and renders readings in three dimensions using React Three Fiber — making dead zones and weak coverage areas immediately visible to ops teams.

Stack: React, Three.js / R3F, Redux Toolkit, Material UI

PROJECTS

Adda – Social Media Web Application *2025*

Full-Stack Personal Project

Architected and built a full-stack social media platform on the MERN stack with a Next.js 16 App Router frontend. Implemented stateless JWT auth via HTTP-only cookies for a decoupled Vercel/Render deployment. Designed a feature-based frontend architecture (features/auth, features/posts, shared/ui) with Redux Toolkit managing optimistic UI and a Zod-validated Express REST API with route-level schema enforcement.

- Implemented two-layer auth — **Next.js Middleware** intercepting unauthenticated routes at the Edge before page render, backed by JWT verification on every Express route via a custom **verifyJWT middleware**; handled cross-domain cookie sharing (sameSite: none) for the decoupled Vercel/Render deployment.
- Built **optimistic UI for social interactions** (likes, comment counts) with Redux Toolkit — updates instantly on action, rolls back silently on API failure; paired with an **IntersectionObserver-based infinite scroll** feed using useCallback and stable dependency arrays to prevent stale closure bugs.
- Designed a scalable **feature-based folder architecture** with Tailwind CSS v4's @theme inline design token system — CSS variables mapped to utility classes so the full light/dark theme swaps from a single :root override with no JavaScript.

Stack: Next.js 16, React, Redux Toolkit, Tailwind CSS v4, Node.js, Express.js, MongoDB, JWT, Zod

GitHub: github.com/akashdeep-majumder/social-media-mern | **Live:** project-adda.vercel.app

Cricket Scoring App *2025*

Personal Project · AI-Assisted

Full-featured cricket tournament manager shipping as a Windows .exe via Electron.js — offline-first with a SQLite-backed database and a real-time scoreboard over Socket.IO.

- Designed end-to-end tournament flow: create tournaments, build team lineups, and score **ball by ball** with a live scoreboard powered by Socket.IO.
- Implemented a **sponsor video upload feature** that plays automatically between scoring intervals — doubling as a lightweight local ad platform for tournament organizers.
- Packaged as a **standalone offline Windows desktop app** via Electron.js; no internet connection required at deployment.

Stack: Electron.js, TypeScript, React, Tailwind CSS, Express.js, SQLite, Socket.IO

GitHub: github.com/akashdeep-majumder/scoring_app

EDUCATION

Tripura University *2019 – 2022*

Bachelor of Science in Physics (Honours)